$$p(y_s \mid \ldots, y_{s-3}, y_{s-1}, y_{s+1}, y_{s+3}, X)$$

$$= p(y_s \mid y_{s-1}, y_{s+1}, X)$$ * By Markov Blanket, $y_s$ depends only on $y_{s-1}, y_{s+1}$

$$\propto \psi_s(y_s, X)\, T(y_{s-1}, y_s)\, T(y_s, y_{s+1})$$

$$p(y_s \mid y_{s-1}, y_{s+1}, X) = \frac{\psi_s(y_s, X)\, T(y_{s-1}, y_s)\, T(y_s, y_{s+1})}{\sum_{y \in \mathcal{Y}} \psi_s(y, X)\, T(y_{s-1}, y)\, T(y, y_{s+1})}$$

Cost of (4), (5) for single iteration of K

1. Ignore the edge potential (*i.e.*, set $T = \mathbf{0}$), which leaves all $y_s$ as independent given $X$.
2. Sample the value of $y_s$ for all $s$ based on such independent distributions.
3. For $k = 1, 2, \ldots, S$ ($S$ being a hyperparameter for tuning, akin to $B$)
4. → Given/fixing $y_1, y_3, y_5, \ldots$, sample for $y_2, y_4, y_6, \ldots$ ($T$ is no longer set to $\mathbf{0}$).
5. → Given/fixing $y_2, y_4, y_6, \ldots$, sample for $y_1, y_3, y_5, \ldots$ ($T$ is no longer set to $\mathbf{0}$).
6. Call the current $\{y_1, y_2, y_3, y_4, \ldots\}$ as a **sample**.
7. Use the $S$ samples to compute the marginal distribution of nodes and edges.

One Gibbs sweep of (4), (5) samples all $m$ nodes

For each node, the conditional of each letter $y \in \mathcal{Y}$ is calculated $|\mathcal{Y}| = 26$, so per-node complexity is $O(|\mathcal{Y}|)$

Therefore, for $m$ nodes, complexity will be $O(m|\mathcal{Y}|)$

For dynamic programming, at each position, for every possible label $\hat{y}_j \in \mathcal{Y}$, we also loop over all previous $\hat{y}_i$ labels, therefore giving $O(|\mathcal{Y}|^2)$

Therefore, the sequence of $m$ nodes gives a complexity of $O(m|\mathcal{Y}|^2)$